PulseIQ Phase 2 Report

Name: Poonam Satish Patole

Track: Data Analysts

Phase: Phase 2

Prepared: 30 October 2026

GitHub repository: PoonamPatole/pulseiq

Git tag: Phase-2

Task status and evidence

Task	Status	Evidence
6. Attendance and leave	Complete locally; corrected monthly rate uses M13 half-day adjustment.	attendance observations
7. Attrition analytics	Complete locally; actual exits and grouped risk are separate; small groups suppressed.	three recorded findings
8. Team productivity	Complete locally; delivery metrics and two manual velocity checks match.	two-team manual verification
9. Payroll cost and reconciliation	Complete locally; figures provisional pending Finance review of five differences and duplicate/orphan records.	five-month reconciliation
10. Budget and KPI targets	Complete locally; variance, KPI status and three hand checks documented.	three manual variance checks

Proof it works

- **Task 6:** Corrected M13 all-department rates: May 2026 92.89%, January 2026 93.29%, April 2026 93.35%; the numerator includes half of unpaid half-days.
- **Task 7:** Largest selected exit rows: Engineering (Mar 2025), 5 / 3.40%; Operations & IT (Mar 2024), 4 / 13.79%; Engineering (Mar 2020), 3 / 5.17%. Tenure uses the SCD row valid at exit.
- **Task 8:** SP-01 checks matched: Analytics & BI, 22 completed tasks / 79 points; Data Platform, 29 / 72.
- **Task 9:** Payroll figures remain provisional pending Finance review. DQ-C01 reports five payroll-to-ledger differences; June 2026 is -INR 18,450; four other months remain under investigation. Duplicate employee/run rows and unmatched IDs also remain unresolved. No rows were excluded.
- **Task 10:** April 2025 hand checks matched: Engineering -INR 1,911,891.25 (-8.19%); Product & Design -INR 461,307.33 (-8.73%); Data & AI -INR 100,184.00 (-1.32%). Negative means underspend.

All five Streamlit apps passed a local AppTest smoke run with no app exceptions; Python compile checks passed. Numeric spot checks are recorded with task evidence.

What I learned

Attendance and leave can have different grains, so leave needs a company-wide caveat. Actual exits and modelled risk answer different questions. Team velocity needs completed-task checks and sprint context. Payroll and budget comparisons need clear sign conventions, traceable sources, and visible uncertainty.

Problems and how I handled them

The attendance feed cannot reliably label long weekends, so holidays are context, not cause. Attrition predictions are grouped and suppress groups below five. Payroll has duplicate employee/run groups, orphan IDs, and five mismatches; I preserved rows and documented differences for review. The answer key is company-level and cannot validate department allocation. Three budget rows were checked by hand.

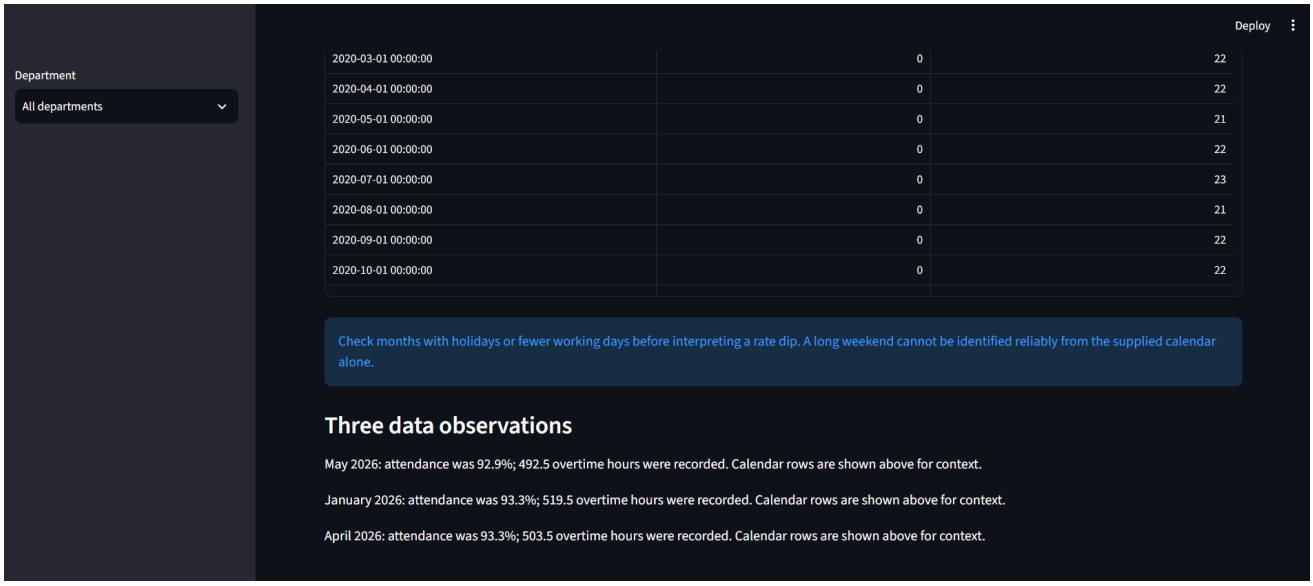
Plan for the next phase

Prepare a shareable deployment, check access from another device, and complete release materials and a demo. Keep payroll exceptions visible until Finance confirms corrections. Commit the final phase files, apply the Phase-2 tag, and record the release reference.

Task screenshots

Task 6. Attendance and leave





Task 7. Attrition analytics

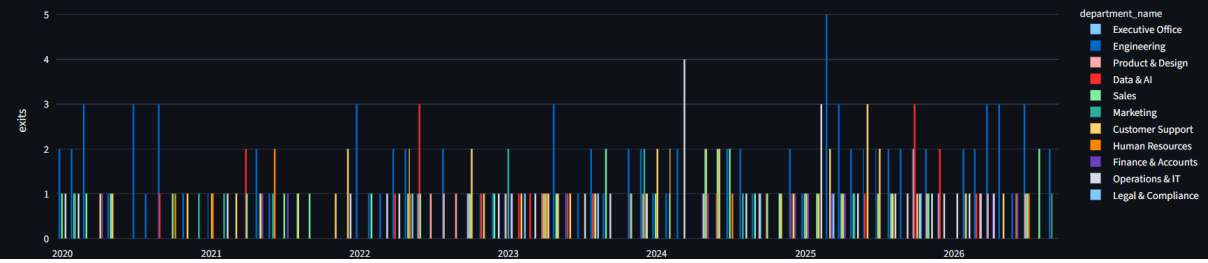
Deploy

Task 7 | Attrition analytics

Actual exits and modelled risk are separate measures. No person-level risk is shown.

Actual exits

Exits by month and department

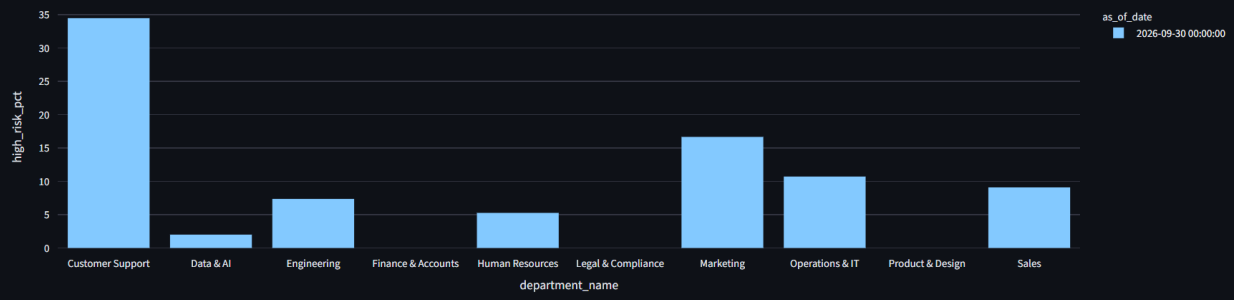


Deploy

Predicted, not certain

Risk is a model estimate for discussion, not a fact, diagnosis, or decision about an employee. Predictions can be wrong and must not be used alone for employment decisions.

High risk share by department (groups of at least 5)



Deploy

D01	Engineering	2026-09-30 00:00:00	163	12	7.362
D08	Finance & Accounts	2026-09-30 00:00:00	16	0	0
D07	Human Resources	2026-09-30 00:00:00	19	1	5.2632
D10	Legal & Compliance	2026-09-30 00:00:00	10	0	0
D05	Marketing	2026-09-30 00:00:00	12	2	16.6667
D09	Operations & IT	2026-09-30 00:00:00	28	3	10.7143
D02	Product & Design	2026-09-30 00:00:00	25	0	0
D04	Sales	2026-09-30 00:00:00	44	4	9.0909

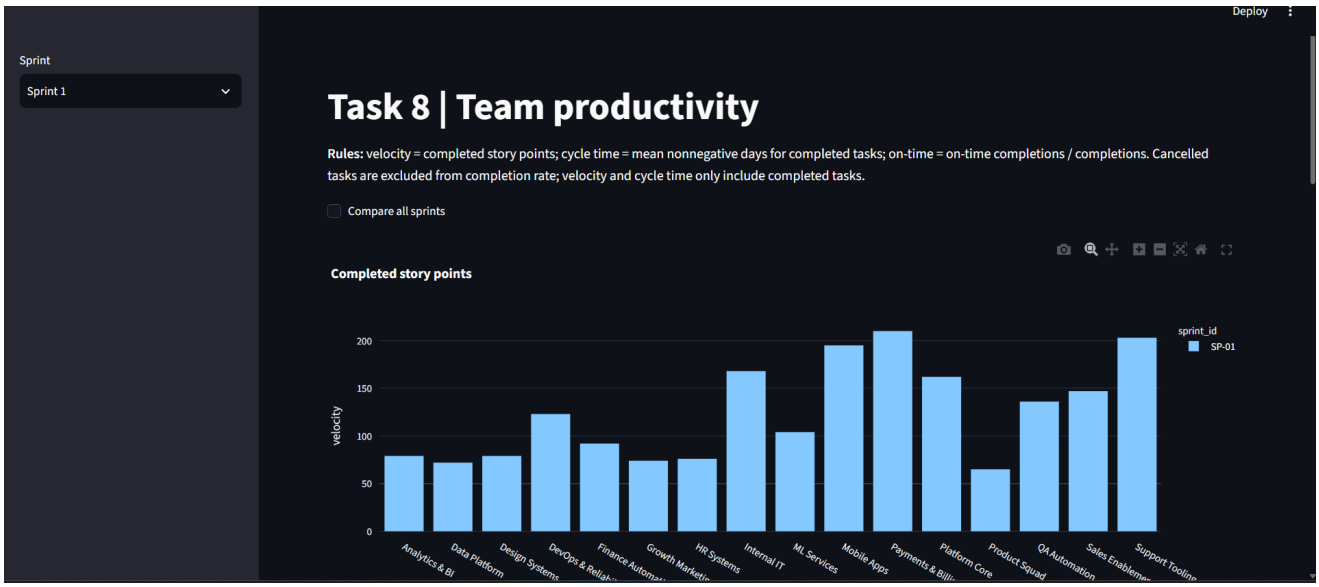
Largest observations

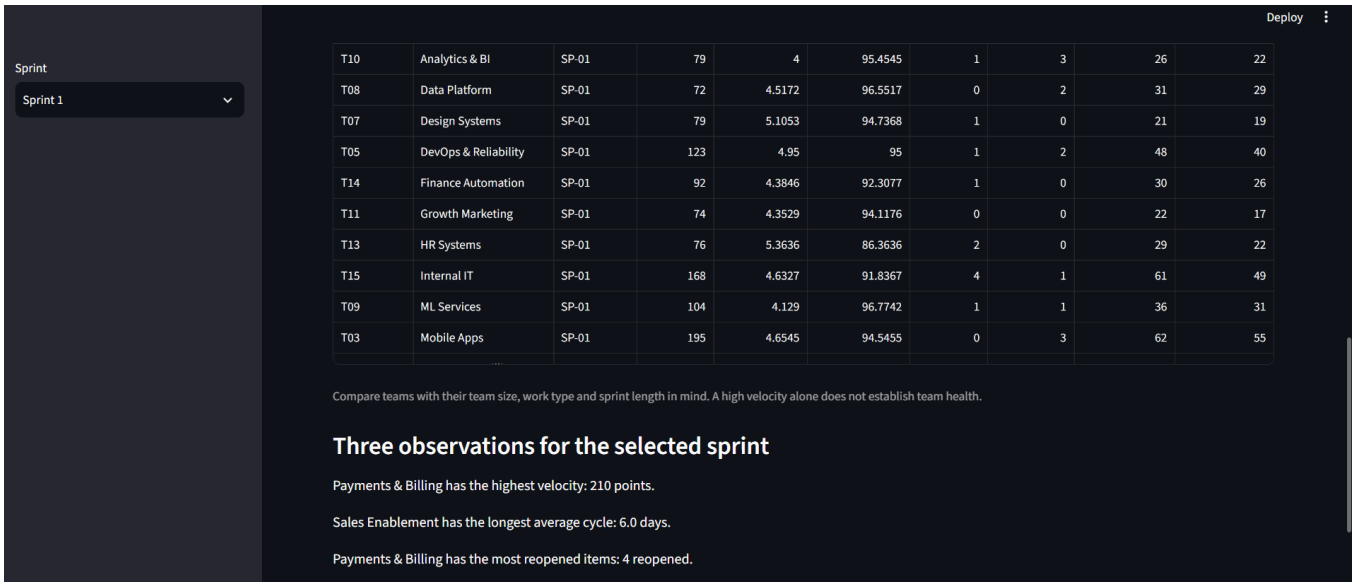
Mar 2025, Engineering: 5 exits (3.40% of starting headcount).

Mar 2024, Operations & IT: 4 exits (13.79% of starting headcount).

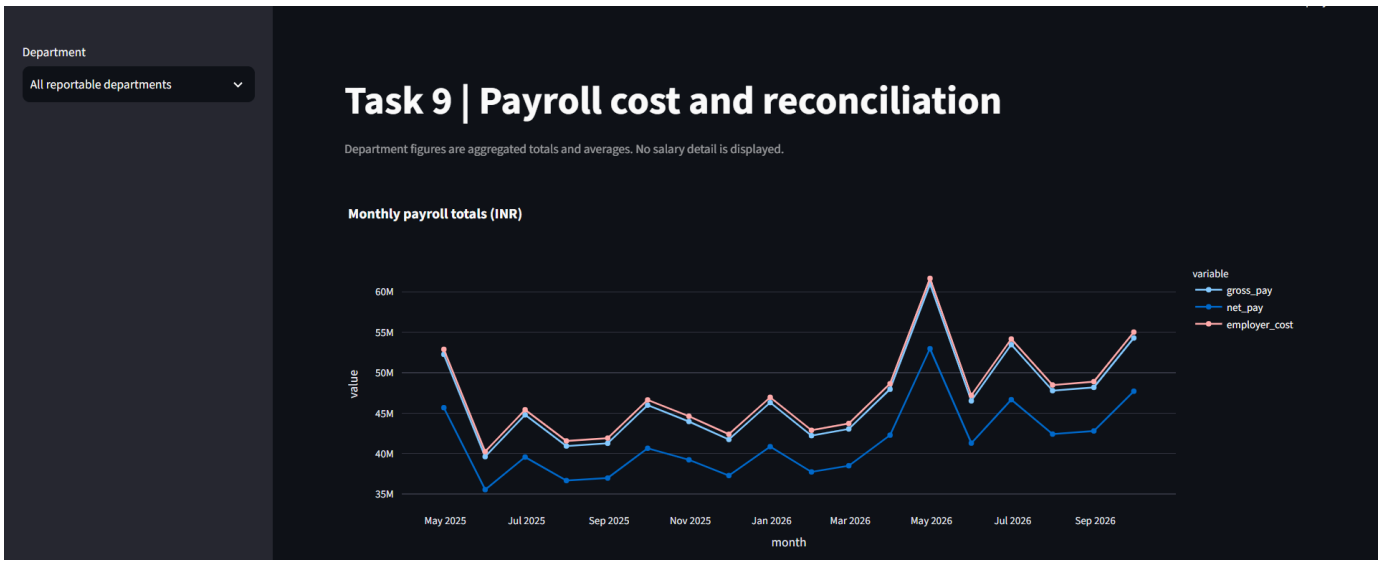
Mar 2020, Engineering: 3 exits (5.17% of starting headcount).

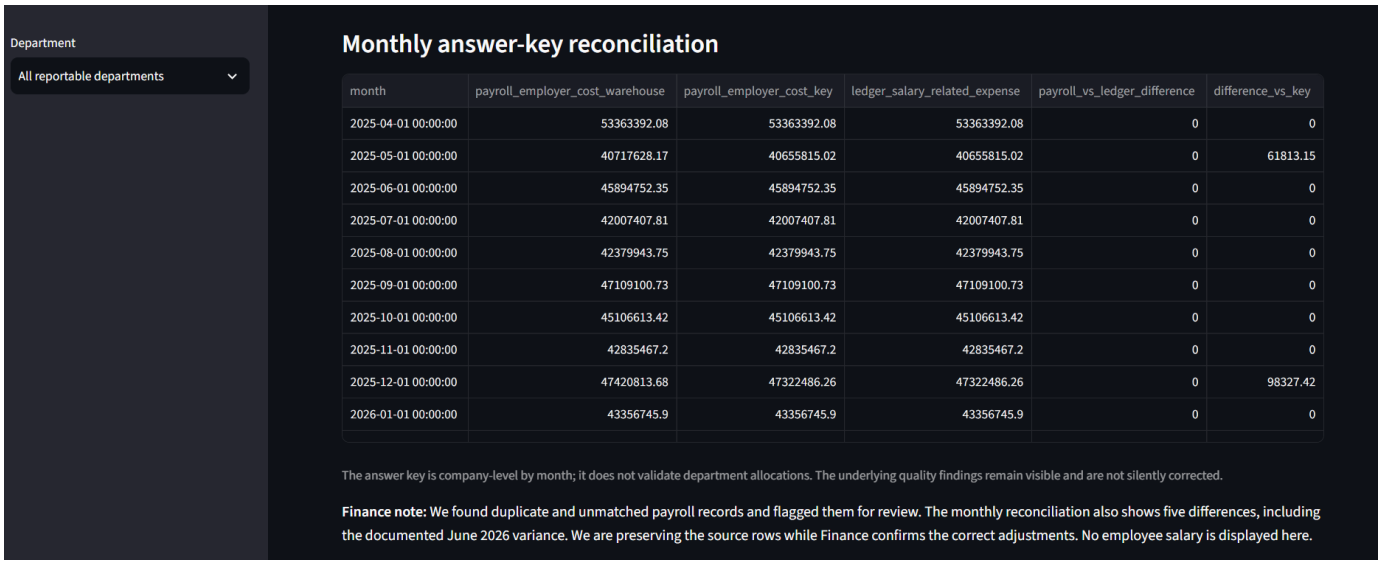
Task 8. Team productivity



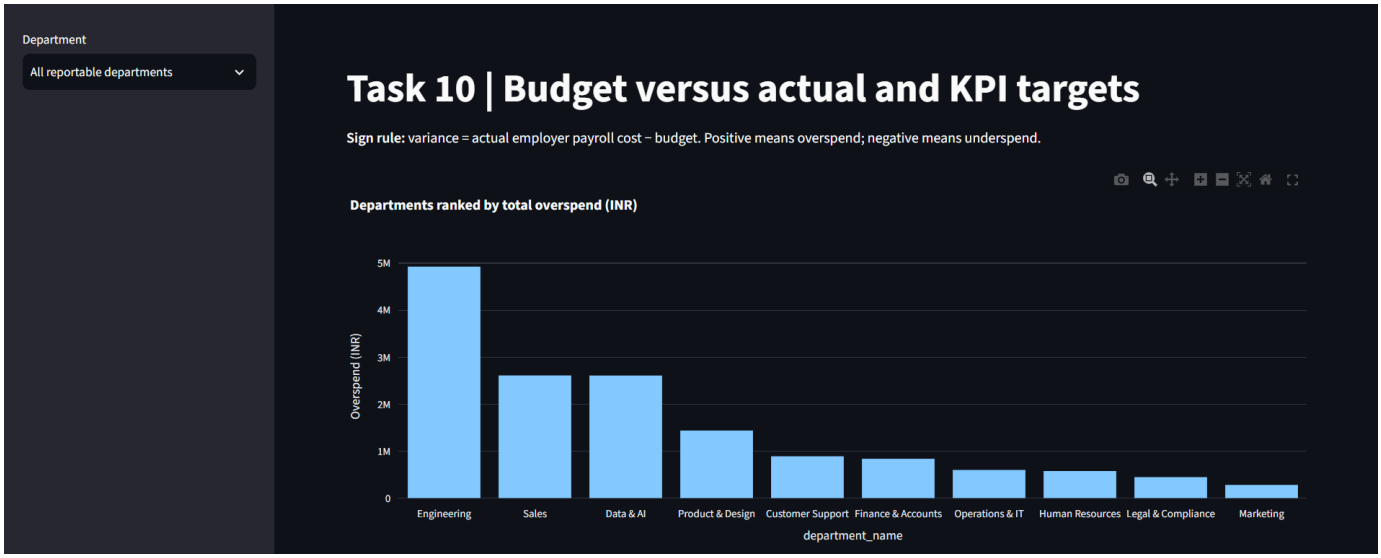


Task 9. Payroll cost and reconciliation





Task 10. Budget and KPI targets



Department

All reportable departments

KPI scorecard

month	department_id	department_name	attendance_rate	attendance_rate_target	attendance_status	overtime_share_pct	overtime_share_target_pct
2025-04-01 00:00:00	D01	Engineering	0.9509	0.94	Green	0	0.5
2025-04-01 00:00:00	D02	Product & Design	0.9598	0.94	Green	0	0.5
2025-04-01 00:00:00	D03	Data & AI	0.9311	0.94	Red	0	0.5
2025-04-01 00:00:00	D04	Sales	0.9353	0.94	Red	0	0.5
2025-04-01 00:00:00	D05	Marketing	0.9116	0.94	Red	0	0.5
2025-04-01 00:00:00	D06	Customer Support	0.952	0.94	Green	1.9669	3
2025-04-01 00:00:00	D07	Human Resources	0.9089	0.94	Red	1.9964	3
2025-04-01 00:00:00	D08	Finance & Accounts	0.9581	0.94	Green	1.417	3
2025-04-01 00:00:00	D09	Operations & IT	0.955	0.94	Green	1.4681	3
2025-04-01 00:00:00	D10	Legal & Compliance	0.9688	0.94	Green	0	0.5

Green means the observed KPI meets its supplied target; red means it misses; no data is not scored. Budget overspend is ranked separately. A missing budget refresh after a hiring freeze should be noted beside the variance and discussed with Finance.

Department

All reportable departments

Budget variance detail

month	department_id	department_name	actual_cost	budget	variance	variance_pct	employee_count	cost_per_employee
2025-04-01 00:00:00	D01	Engineering	21428108.75	23340000	-1911891.25	-8.1915	145	147780.0603
2025-04-01 00:00:00	D02	Product & Design	4823692.67	5285000	-461307.33	-8.7286	23	209725.7683
2025-04-01 00:00:00	D03	Data & AI	7461816	7562000	-100184	-1.3248	49	152281.9592
2025-04-01 00:00:00	D04	Sales	5885879.82	5481000	404879.82	7.387	39	150919.9954
2025-04-01 00:00:00	D05	Marketing	1669788.68	1762000	-92211.32	-5.2333	11	151798.9709
2025-04-01 00:00:00	D06	Customer Support	3070909.23	3191000	-120090.77	-3.7634	50	61418.1846
2025-04-01 00:00:00	D07	Human Resources	1652543.06	1655000	-2456.94	-0.1485	14	118038.79
2025-04-01 00:00:00	D08	Finance & Accounts	2620991.39	2753000	-132008.61	-4.7951	16	163811.9619
2025-04-01 00:00:00	D09	Operations & IT	2751496.9	2887000	-135503.1	-4.6936	30	91716.5633
2025-04-01 00:00:00	D10	Legal & Compliance	1517356.58	1473000	44356.58	3.0113	8	189669.5725

Deploy

Department

All reportable departments

2025-04-01 00:00:00	D10	Legal & Compliance	0.9688	0.94	Green	0	0.5

Green means the observed KPI meets its supplied target; red means it misses; no data is not scored. Budget overspend is ranked separately. A missing budget refresh after a hiring freeze should be noted beside the variance and discussed with Finance.

Top two overspend recommendations

Engineering: review hiring, vacancies and payroll changes against the approved plan; cumulative overspend is INR 4,926,386. Confirm whether the plan reflects the hiring-freeze date before changing staffing.

Sales: review hiring, vacancies and payroll changes against the approved plan; cumulative overspend is INR 2,609,956. Confirm whether the plan reflects the hiring-freeze date before changing staffing.

Three manual checks

Choose three department-month rows and verify actual, budget, actual − budget, and percentage = variance ÷ budget × 100. Record the results in evidence/verification.md.